cytokine blockade has been demonstrated to be effective in preventing immune activation in some diseases or in inhibiting the extensive inflammatory effector mechanisms characteristic of these diseases. New therapies have also been developed to target lymphoid cells more specifically by blocking a costimulatory signal needed for T or B cell activation, by blocking the migratory capacity of lymphocytes, or by eliminating the effector T cells or B cells. The efficacy of these therapies in some diseases—e.g., SLE (belimumab), rheumatoid arthritis (TNF neutralization, IL-6 receptor blockade, CD28 competition, B cell depletion, IL-1 competition), psoriasis (IL-12/23 depletion, TNF neutralization), and inflammatory bowel disease (TNF neutralization, IL-12/23 neutralization)—has been demonstrated. One major advance in inhibiting effector mechanisms has been the introduction of cytokine blockade that appears to limit organ damage in some diseases, including rheumatoid arthritis, inflammatory bowel disease, psoriasis, and the spondyloarthritides. Small molecules that block cytokine signaling pathways by blocking the Janus kinase (JAK) family of kinases have recently been introduced into the clinic. Biologicals that delete B cells (anti-CD20 antibody) have recently been approved for the treatment of rheumatoid arthritis and have demonstrated efficacy in other autoimmune diseases as well. Their efficacy in diseases characterized by pathogenic effector T cells has highlighted the importance of B cells as antigen presenting cells in autoimmune diseases. Finally, there is renewed interest in cellular therapies in autoimmune diseases, including hematopoietic stem cell reconstitutions and treatment with immunosuppressive mesenchymal stem cells. Therapies that prevent target-organ damage or support target-organ function also remain important in the management of autoimmune disease.

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349 Systemic Lupus Erythematosus

Bevra Hannahs Hahn

DEFINITION AND PREVALENCE

Systemic lupus erythematosus (SLE) is an autoimmune disease in which organs and cells undergo damage initially mediated by tissue-binding autoantibodies and immune complexes. In most patients, autoantibodies are present for a few years before the first clinical symptom appears. Ninety percent of patients are women of child-bearing years; people of all genders, ages, and ethnic groups are susceptible. Prevalence of SLE in the United States is 20–150 per 100,000 women depending on race and gender; highest prevalence is in African-American and Afro-Caribbean women, and lowest prevalence is in white men.

PATHOGENESIS AND ETIOLOGY

The proposed pathogenic mechanisms of SLE are illustrated in Fig. 349-1. The abnormal immune responses underlying SLE may be summarized as leading to production of increased quantities and immunogenic forms of nucleic acids, their accompanying proteins, and other self-antigens. The process may begin with autoimmunity-inducing activation of innate immunity, partly through binding of DNA/RNA/proteins by toll-like receptors in those cells. The changes include dendritic cells producing interferon α (IFNα), activated macrophages producing inflammatory cytokines/chemokines such as interleukin (IL12), tumor necrosis factor α (TNFα), and the B cell maturation/survival factor BLys/BAFF, neutrophils releasing DNA/protein-containing nets, and natural killer (NK) cells unable to kill autoreactive T and B cells or to produce the transforming growth factor β (TGFβ) needed for development of regulatory T cells. Upregulation of genes induced by IFNs is a genetic "signature" in peripheral blood cells of 50–80% of SLE patients. The innate immune system interacts with the B and T cells of adaptive immunity, which further drive autoimmune responses. T lymphocytes have altered metabolism (abnormal mitochondrial electron transport, membrane potential, and oxidative stress), increased glucose utilization, increased pyruvate production, activation of mTOR, and increased autophagy. T and B cells are more easily activated and driven into apoptosis than are normal cells, probably due to autoantibodies binding them plus abnormal signaling after engagement of surface molecules resulting in abnormally low production of IL2, which is required for T cell survival. B cells present antigen and secrete IL6 and IL10, further promoting autoreactive B cell survival (which is also favored by estrogen). Lupus phagocytic cells have reduced capacity to clear immune complexes, apoptotic cells, and their DNA/RNA/Ro/La and phospholipid containing surface blebs. The result is persistence of large quantities of autoantigens and resultant large quantities of autoantibodies with increased numbers of activated B cells and plasmablasts/plasma cells, and autoreactive T cells with shifts away from regulatory populations toward increased numbers and functions of Th1, T17, and Tfh cells, all of which promote production of autoantibodies and tissue damage. This damage begins with deposition of autoantibodies and/or immune complexes, followed by destruction mediated by complement activation and release of cytokines/chemokines. Non-immune tissue-fixed cells are then activated to produce more inflammation and damage, such as basal cells of the dermis, synovial fibroblasts, renal mesangial cells, podocytes and tubular epithelium, and endothelial cells throughout the body. Meanwhile, the initial immune attack is attracting into the target tissues additional B and T cells, monocytes/macrophages, dendritic cells, and plasma cells. Inflammation also causes release of vasoactive peptides, oxidative damage, growth factors and fibrosing factors. Sclerosis/fibrosis with irreversible tissue damage can occur in multiple tissues including kidneys, lungs, blood vessels, and skin. Each of these processes depends on the individual's genetic background, environmental influences, and epigenetics. Autoantibodies of SLE are referred to in Fig. 349-1 and described in Table 349-1.

SLE is a multigenic disease. Rare single-gene defects confer high hazard ratios (HRs) for SLE (5–25), including homozygous deficiencies of early components of complement (C1q,r,s; C2; C4) and a mutation in *TREX1* (encoding a DNAase) on the X chromosome. In most genetically susceptible individuals, normal alleles of multiple genes each contribute a small amount to abnormal immune/inflammation/tissue damage responses; if enough predisposing variations are present, disease results. Approximately 60 genes with alleles increasing risk for SLE and/or lupus nephritis (examples listed in Fig. 349-1 which includes most with HR ≥1.5) have been identified in recent genome-wide association studies in different racial groups. Individually, they confer an HR for SLE of 1.5–3 and even in combination account for only 18% of disease susceptibility, suggesting that environmental exposures and epigenetics play major roles. Predisposing, antigen-presenting human leukocyte antigen (HLA) molecules are most commonly found, in multiple ethnic groups (HLA DRB1 *0301 and *1501 and DR3), as well as multiple genes across the major histocompatibility complex (MHC)

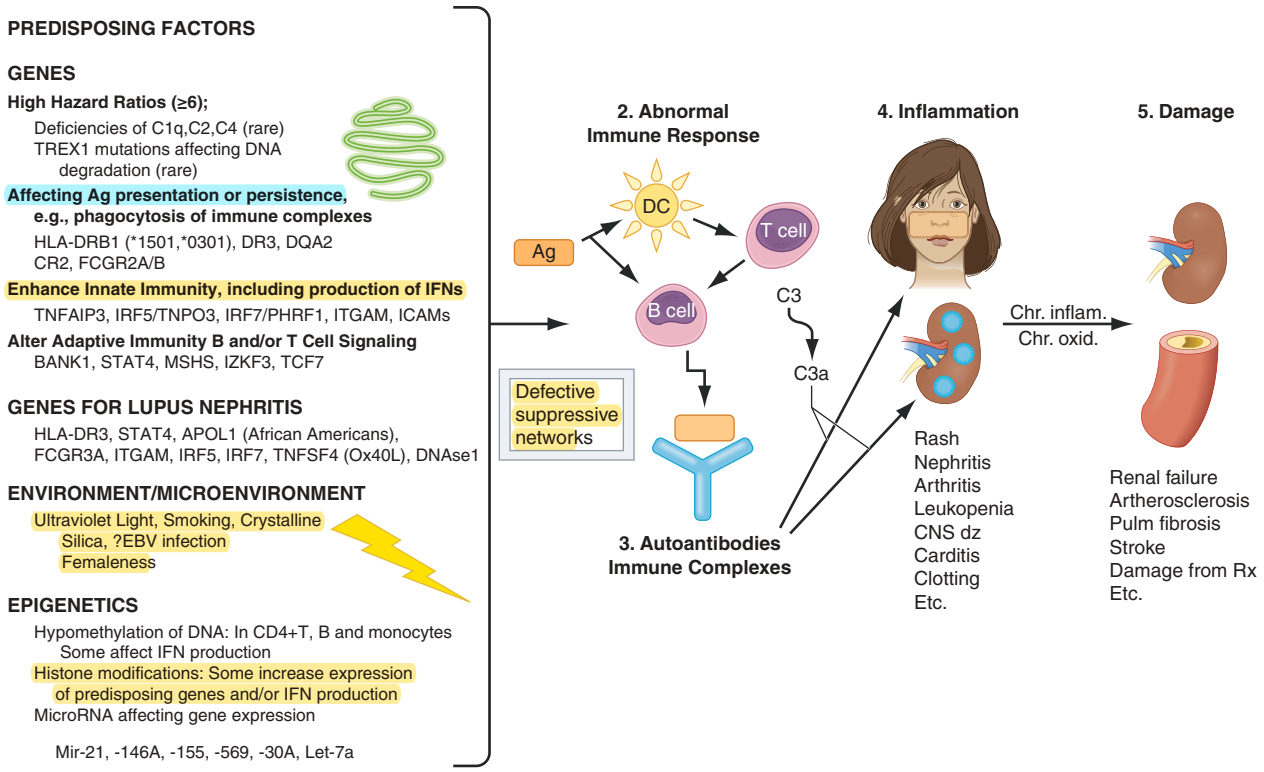


FIGURE 349-1 Pathogenesis of systemic lupus erythematosus (SLE). Pathogenesis is related in large part to production of increased quantities and immunogenic forms of nucleic acids and other self-antigens, which drive autoimmune-inducing activation of innate immunity, autoantibodies, and T cells. Interactions between genes, environment, and epigenetic changes drive increased autophagy, Ag presentation, neutrophil netosis, autoantibody formation with increased plasma cells, and production of pathogenic effector T cells in Th1, Th17, and Tfh subsets, with ineffective regulatory networks. Genes confirmed in more than one genome-wide association analysis in multiple racial groups that increase susceptibility to SLE or lupus nephritis (HR ≥1.5) are listed (reviewed in *Teruel M, Alarcon-Riquelme ME: The genetic basis of systemic lupus erythematosus: What are the risk factors and what have we learned. J Autoimmun 74:161, 2016; and Iwamoto T, Niewold TB: Genetics of human lupus nephritis. Clin Immunol 2016. Epub ahead of print*). Epigenetics are reviewed in *Long H et al: The critical role of epigenetics in systemic lupus erythematosus and autoimmunity, J Autoimmun 2016. Epub ahead of print.* Gene-environment interactions (reviewed in *Barbhaiya M, Costenbader KH: Environmental exposures and the development of systemic lupus erythematosus. Curr Opin Rheumatol 2016. Epub ahead of print*) result in abnormal immune responses that generate pathogenic autoantibodies and immune complexes that deposit in tissue, activate complement, induce cytokine and chemokine release causing inflammation, and over time lead to irreversible organ damage (reviewed in *Anders HJ, Rovin B: A pathophysiology-based approach to the diagnosis and treatment of lupus nephritis. Kidney Int 90:493, 2016;* and *Hahn BH: Pathogenesis of SLE, in Dubois Lupus Erythematosus, 8th ed, DJ Wallace, BH Hahn, (eds). Philadelphia, Elsevier, 2013*). Ag, antigen; C1q, complement system; C3, complement component; CNS, central nervous system; DC, dendritic cell; EBV, Epstein-Barr virus; HLA, human leukocyte antigen; FcR, immunoglobulin Fc-binding receptor; IL, interleukin; MCP, monocyte chemotactic protein; PTPN, phosphotyrosine phosphatase; UV, ultraviolet.

120-gene region. Non-HLA genetic factors are listed in Fig. 349-1 and include polymorphisms that affect innate and adaptive immunity pathways. Note the large number that influences IFN production—the most characteristic gene expression pattern of SLE patients. Other genes affect clearance of apoptotic cells or immune complexes, influence neutrophil adherence (*ITGAM*), and DNA repair (*TREX-1*). Some polymorphisms influence clinical manifestations; such as single nucleotide polymorphisms (SNPs) of *STAT4* that associate with severe disease, anti-DNA, nephritis, and antiphospholipid syndrome (APS), and an allele of *FCGRIIA* encoding a receptor that binds immune complexes poorly and predisposes to nephritis. Some gene effects are in promoter regions (e.g., IL-10), and others are conferred by copy numbers (e.g., C4A, TLR7). In addition, multiple epigenetic changes characterize SLE, including hypomethylation of DNA in CD4+ T cells, B cells, and monocytes, including genes that control production of type 1 interferons, and histone modifications. Some of these changes are mediated by microRNAs associated with SLE including some that control DNA Methyl transferases (DNMTs), such as mIR-146a, that control methylation of DNA in CD4+ T cells and IFN production. Some gene polymorphisms contribute to several autoimmune diseases, such as *STAT4* and *CTLA4*. All of these gene polymorphisms/transcription/epigenetic combinations influence immune responses to the external and internal environment; when such responses are too high and/or too prolonged and/or inadequately regulated, autoimmune disease is favored.

Female sex is permissive for SLE with evidence for hormone effects, genes on the X chromosome, and epigenetic differences between genders playing a role. Females of many mammalian species make higher antibody responses than males. Women exposed to estrogen-containing oral contraceptives or hormone replacement have an increased risk of developing SLE (HR 1.2–2). Estradiol binds to receptors on T and B lymphocytes, increasing activation and survival of those cells, especially autoreactive subsets, thus favoring prolonged immune responses. Genes on the X chromosome that influence SLE, such as *TREX-1*, may play a role in gender predisposition, possibly because some genes on the second X in females are not silent. People with XXY karyotype (Klinefelter's syndrome) have a significantly increased risk for SLE.

Several environmental stimuli may influence SLE (Fig. 349-1). Exposure to ultraviolet light causes flares of SLE in ~70% of patients, possibly by increasing apoptosis in skin cells or by altering DNA and intracellular proteins to make them antigenic. Some infections and lupus-inducing drugs activate autoreactive T and B cells; if such cells are not appropriately regulated, prolonged autoantibody production occurs. Most SLE patients have autoantibodies for 3 years or more before the first symptoms of disease, suggesting that regulation controls the degree of autoimmunity for years before quantities and qualities of autoantibodies, pathogenic B and T cells, and activated tissue-fixed cells such as macrophages cause clinical disease. Epstein-Barr virus (EBV) may be one infectious agent that can trigger SLE in susceptible individuals. Children and adults with SLE are more likely to be infected by EBV than age-, sex-, and ethnicity-matched controls. EBV contains amino acid sequences that mimic sequences on human spliceosomes (RNA/protein antigens) often recognized by autoantibodies in people

TABLE 349-1 Autoantibodies in Systemic Lupus Erythematosus (SLE)

ANTIBODY	PREVALENCE, %	ANTIGEN RECOGNIZED	CLINICAL UTILITY
Antinuclear antibodies	98	Multiple nuclear	Best screening test; repeated negative tests by immunofluorescence make SLE unlikely
Anti-dsDNA	70	DNA (double-stranded)	High titers are SLE-specific and in some patients correlate with disease activity, nephritis, vasculitis. Crithidia immunofluorescence is more specific for SLE than ELISA methods.
Anti-Sm	25	Protein complexed to 6 species of nuclear U1 RNA	Specific for SLE; no definite clinical correlations; most patients also have anti-RNP; more common in blacks and Asians than whites
Anti-RNP	40	Protein complexed to U1 RNA	Not specific for SLE; high titers associated with syndromes that have overlap features of several rheumatic syndromes including SLE; more common in blacks than whites; correlates with high IFN-induced gene signature
Anti-Ro (SS-A)	30	Protein complexed to hY RNA, primarily 60 kDa and 52 kDa	Not specific for SLE; associated with sicca syndrome, predisposes to subacute cutaneous lupus, and to neonatal lupus with congenital heart block; associated with decreased risk for nephritis
Anti-La (SS-B)	10	47-kDa protein complexed to hY RNA	Usually associated with anti-Ro; associated with decreased risk for nephritis
Antihistone	70	Histones associated with DNA (in nucleosome, chromatin)	More frequent in drug-induced lupus than in SLE
Antiphospholipid	50	Phospholipids, β_2 glycoprotein 1 (β_2G1) cofactor, prothrombin	Three tests available—ELISAs for cardiolipin and β_2G1, sensitive prothrombin time (DRVVT) for lupus anticoagulant; predisposes to clotting, fetal loss, thrombocytopenia
Antierythrocyte	60	Erythrocyte membrane	Measured as direct Coombs test; a small proportion develops overt hemolysis
Antiplatelet	30	Surface and altered cytoplasmic antigens on platelets	Associated with thrombocytopenia, but sensitivity and specificity are not good; this is not a useful clinical test
Antineuronal (includes antiglutamate receptor 2)	60	Neuronal and lymphocyte surface antigens	In some series, a positive test in CSF correlates with active CNS lupus
Antiribosomal P	20	Protein in ribosomes	In some series, a positive test in serum correlates with depression or psychosis due to CNS lupus

Abbreviations: CNS, central nervous system; CSF, cerebrospinal fluid; DRVVT, dilute Russell viper venom time; ELISA, enzyme-linked immunosorbent assay.

with SLE. Current tobacco smoking increases risk for SLE (HR 1.5). Prolonged occupational exposure to crystalline silica (e.g., inhalation of soap powder dust or soil in farming activities) increases risk (HR 4.3) in African-American women. Drinking alcohol (2 glasses of wine a week or ½ of an alcoholic drink daily) reduces the risk of SLE. Thus, interplay between genetic susceptibility, environment, gender, race, and abnormal immune responses results in autoimmunity **(Chap. 348)**.

PATHOLOGY

In SLE, biopsies of affected skin show deposition of Ig at the dermal-epidermal junction (DEJ), injury to basal keratinocytes, and inflammation dominated by T lymphocytes in the DEJ and around blood vessels and dermal appendages. Clinically unaffected skin may also show Ig deposition at the DEJ. These patterns are not specific for dermatologic SLE; however, they are highly suggestive.

In renal biopsies, the pattern and severity of injury are important in diagnosis and in selecting the best therapy. Most recent clinical studies of lupus nephritis have used the International Society of Nephrology (ISN) and the Renal Pathology Society (RPS) classification **(Table 349-2)**. In the ISN/RPS classification, the addition of "a" for active and "c" for chronic changes gives physicians information regarding the potential reversibility of disease. The system focuses on glomerular disease, although the presence of tubular interstitial and vascular disease, as well as the chronicity score in both glomeruli and interstitium, are important in predicting clinical outcomes. In general, class III and IV disease, as well as class V accompanied by III or IV disease, should be treated with aggressive immunosuppression if possible, because there is a high risk for end-stage renal disease (ESRD) if patients are untreated or undertreated. In contrast, treatment for lupus nephritis is not recommended in patients with class I or II disease or with extensive irreversible changes. In the recent Systemic Lupus International Collaborating Clinic (SLICC) criteria for classification of SLE, a diagnosis can be established on the basis of renal histology in the presence of lupus autoantibodies, without meeting additional criteria totaling 4 **(Table 349-3)**.

Histologic abnormalities in blood vessels may also determine therapy. Patterns of vasculitis are not specific for SLE but may indicate active disease: leukocytoclastic vasculitis is most common **(Chap. 356)**.

Lymph node biopsies are usually performed to rule out infection or malignancies. In SLE, they show nonspecific diffuse chronic inflammation.

DIAGNOSIS

The diagnosis of SLE is based on characteristic clinical features and autoantibodies. Current criteria for classification are listed in Table 349-3, and an algorithm for diagnosis and initial therapy is shown in **Fig. 349-2**. The criteria are intended for confirming the diagnosis of SLE in patients included in studies; the author uses them in individual patients for estimating the probability that a disease is SLE. Any combination of four or more criteria, with at least one in the clinical and one in the immunologic category, well documented at any time during an individual's history, makes it likely that the patient has SLE. (Specificity and sensitivity are ~93% and ~92%, respectively.) In many patients, criteria accrue over time. Antinuclear antibodies (ANA) are positive in >98% of patients during the course of disease; repeated negative tests by immunofluorescent methods suggest that the diagnosis is not SLE, unless other autoantibodies are present (Fig. 349-2). High-titer IgG antibodies to double-stranded DNA and antibodies to the Sm antigen are both specific for SLE and, therefore, favor the diagnosis in the presence of compatible clinical manifestations. The presence in an individual of multiple autoantibodies without clinical symptoms should not be considered diagnostic for SLE, although such persons are at increased risk.

INTERPRETATION OF CLINICAL MANIFESTATIONS

When a diagnosis of SLE is made, it is important to establish the severity and potential reversibility of the illness and to estimate the possible consequences of various therapeutic interventions. In the following paragraphs, descriptions of some disease manifestations begin with relatively mild problems and progress to those more life-threatening.

TABLE 349-2 Classification of Lupus Nephritis (International Society of Nephrology and Renal Pathology Society)

Class I: Minimal Mesangial Lupus Nephritis

Normal glomeruli by light microscopy, but mesangial immune deposits by immunofluorescence.

Class II: Mesangial Proliferative Lupus Nephritis

Purely mesangial hypercellularity of any degree or mesangial matrix expansion by light microscopy, with mesangial immune deposits. A few isolated subepithelial or subendothelial deposits may be visible by immunofluorescence or electron microscopy, but not by light microscopy.

Class III: Focal Lupus Nephritis

Active or inactive focal, segmental or global endo- or extracapillary glomerulonephritis involving ≤50% of all glomeruli, typically with focal subendothelial immune deposits, with or without mesangial alterations.

- Class III (A): Active lesions—focal proliferative lupus nephritis
- Class III (A/C): Active and chronic lesions—focal proliferative and sclerosing lupus nephritis
- Class III (C): Chronic inactive lesions with glomerular scars—focal sclerosing lupus nephritis

Class IV: Diffuse Lupus Nephritis

Active or inactive diffuse, segmental or global endo- or extracapillary glomerulonephritis involving ≥50% of all glomeruli, typically with diffuse subendothelial immune deposits, with or without mesangial alterations. This class is divided into diffuse segmental (IV-S) lupus nephritis when ≥50% of the involved glomeruli have segmental lesions, and diffuse global (IV-G) lupus nephritis when ≥50% of the involved glomeruli have global lesions. Segmental is defined as a glomerular lesion that involves less than one-half of the glomerular tuft. This class includes cases with diffuse wire loop deposits but with little or no glomerular proliferation.

- Class IV-S (A): Active lesions—diffuse segmental proliferative lupus nephritis
- Class IV-G (A): Active lesions—diffuse global proliferative lupus nephritis
- Class IV-S (A/C): Active and chronic lesions—diffuse segmental proliferative and sclerosing lupus nephritis
- Class IV-G (A/C): Active and chronic lesions—diffuse global proliferative and sclerosing lupus nephritis
- Class IV-S (C): Chronic inactive lesions with scars—diffuse segmental sclerosing lupus nephritis
- Class IV-G (C): Chronic inactive lesions with scars—diffuse global sclerosing lupus nephritis

Class V: Membranous Lupus Nephritis

Global or segmental subepithelial immune deposits or their morphologic sequelae by light microscopy and by immunofluorescence or electron microscopy, with or without mesangial alterations. Class V lupus nephritis may occur in combination with class III or IV, in which case both will be diagnosed. Class V lupus nephritis may show advanced sclerosis.

Class VI: Advanced Sclerotic Lupus Nephritis

≥90% of glomeruli globally sclerosed without residual activity.

Note: Indicate and grade (mild, moderate, severe) tubular atrophy, interstitial inflammation and fibrosis, and severity of arteriosclerosis or other vascular lesions.

Source: JJ Weening et al: Kidney Int 65:521, 2004. Reprinted by permission from Macmillan Publishers Ltd., Copyright 2004.

TABLE 349-3 Systemic Lupus International Collaborating Clinic Criteria for Classification of Systemic Lupus Erythematosus

CLINICAL MANIFESTATIONS	IMMUNOLOGIC MANIFESTATIONS
Skin	ANA > reference negative value
Acute, subacute cutaneous LE (photosensitive, malar, maculopapular, bullous)	Anti-dsDNA >reference, if by ELISA 2x reference
Chronic cutaneous LE (discoid lupus, panniculitis, lichen planus-like, hypertrophic verrucous, chillblains)	Anti-Sm
Oral or nasal ulcers	Antiphospholipid (any of lupus anticoagulant, false-positive RPR, anti-cardiolipin, anti-β glycoprotein I
Nonscarring Alopecia	Low serum complement (C3, C4 or CH50)
Synovitis involving ≥2 joints	Positive direct Coombs test in absence of hemolytic anemia
Serositis (pleurisy, pericarditis)	
Renal	
Prot/Cr ≥0.5	
RBC casts	
Biopsy[a]	
Neurologic	
Seizures, psychosis, mononeuritis, myelitis, peripheral or cranial neuropathies, acute confusional state	
Hemolytic anemia	
Leukopenia (<4000/μL) or Lymphopenia (<1000/μL)	
Thrombocytopenia (<100,000/μL)	

[a]Renal biopsy read as systemic lupus qualifies for classification as SLE if any lupus autoantibodies are present, even if total criteria are fewer than 4.

Interpretation: Presence of any four criteria (must have at least 1 in each category) qualifies patient to be classified as having SLE with 93% specificity and 92% sensitivity. American College of Rheumatology is developing new criteria for SLE. For update see website *Rheumatology.org*.

Abbreviations: ANA, antinuclear antibody; Cr, creatinine; LE, lupus erythematosus; Prot, protein.

Source: M Petri et al: Arthritis Rheum 64:2677, 2012. Because these criteria are relatively new, some currently ongoing clinical studies use prior American College of Rheumatology Criteria; see EM Tan et al: Arthritis Rheum 25:1271, 1982; update MC Hochberg: Arthritis Rheum 40:1725, 1997.

OVERVIEW AND SYSTEMIC MANIFESTATIONS

At its onset, SLE may involve one or several organ systems; over time, additional manifestations may occur (Tables 349-3 and 349-4). Most of the autoantibodies characteristic of each person are present at the time clinical manifestations appear (Tables 349-1 and 349-3). Severity of SLE varies from mild and intermittent to severe and fulminant. Approximately 85% of patients have either continuing active disease (on current treatment) or one or more flares of active disease annually. Permanent complete remissions (absence of symptoms with no treatment) are rare; however, low-level disease activity on treatments such as hydroxychloroquine and/or low dose prednisone is achievable in ~35% of patients. Systemic symptoms, particularly fatigue and myalgias/arthralgias, are present most of the time. Severe systemic illness requiring high dose glucocorticoid therapy can occur with fever, prostration, weight loss, and anemia with or without other organ-targeted manifestations.

MUSCULOSKELETAL MANIFESTATIONS

Most people with SLE have intermittent polyarthritis, varying from mild to disabling, characterized by soft tissue swelling and tenderness in joints and/or tendons, most commonly in hands, wrists, and knees. Joint deformities (hands and feet) develop in only 10%. Erosions on joint x-rays are rare but can be identified by ultrasound in 10–50% of patients. Some individuals have rheumatoid-like arthritis with erosions and fulfill criteria for both RA and SLE ("rhupus"). Joint pain is the most common reason that patients increase their dose of glucocorticoids. If pain persists in a single joint, such as knee, shoulder, or hip, a diagnosis of ischemic necrosis of bone (INB) should be considered, particularly if there are no other manifestations of active SLE, because INB prevalence is increased in SLE, especially in patients treated with systemic glucocorticoids. Myositis with clinical muscle weakness, elevated creatine kinase levels, positive magnetic resonance imaging (MRI) scan, and muscle necrosis and inflammation on biopsy can occur, although most patients have myalgias without frank myositis. Glucocorticoid therapies (commonly) and antimalarial therapies (rarely) can cause muscle weakness; these adverse effects must be distinguished from active inflammatory disease.

CUTANEOUS MANIFESTATIONS

Lupus dermatitis can be classified as acute, subacute, or chronic, and there are many different types of lesions encompassed within these groups. Discoid lupus erythematosus (DLE) is the most common chronic dermatitis in lupus; lesions are roughly circular with slightly raised, scaly hyperpigmented erythematous rims and depigmented, atrophic centers in which all dermal appendages are permanently destroyed. Lesions can be disfiguring, particularly on the face and

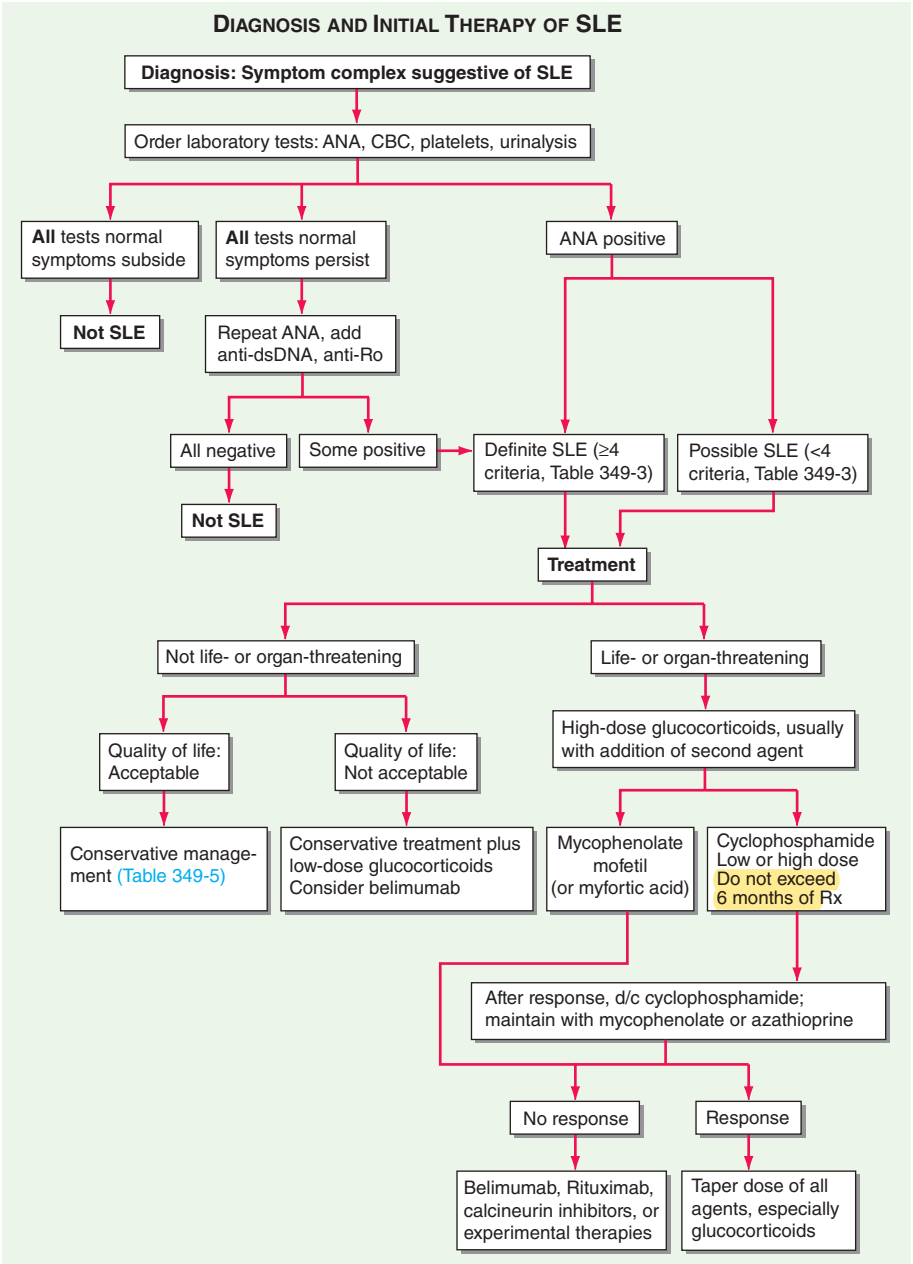


FIGURE 349-2 Algorithm for diagnosis and initial therapy of systemic lupus erythematosus (SLE). For guidelines on management of lupus and lupus nephritis, see Hahn BH et al: Arthritis Care Res (Hoboken) 64:797, 2012; Bertsias GK et al: Ann Rheum Dis 71:1771, 2012; Anders HJ, Rovin B: Kidney Int 2016 Epub ahead of print. For details on mycophenolate and cyclophosphamide induction and maintenance therapies, see Henderson L et al: Cochrane Database Syst Rev 12:CD002922, 2012; Ginzler EM et al: Arthritis Rheum 62:211, 2010; Houssiau FA et al: Ann Rheum Dis 69:61, 2010; and Dooley MA et al: N Engl J Med 365:1886, 2011. For belimumab in treatment, see Bruce IN et al: Long term organ damage accumulation and safety in patients with SLE treated with belimumab plus standard of care. Lupus 25:699, 2016; Hahn BH: N Eng J Med 368:1528, 2013. For rituximab, see Lightstone L: Lupus 22:390, 2013 and Rovin BH et al: Arthritis Rheum 64:1215, 2012. For tacrolimus, see Liu Z et al: Multitarget therapy for induction treatment of lupus nephritis: A randomized trial. Ann Intern Med 162:18, 2015. ANA, antinuclear antibodies; CBC, complete blood count.

scalp. Treatment consists primarily of topical or locally injected glucocorticoids and systemic antimalarials. Only 5% of people with DLE have SLE (although half have positive ANA); however, among individuals with SLE, as many as 20% have DLE. The most common acute SLE rash is a photosensitive, slightly raised erythema, occasionally scaly, on the face (particularly the cheeks and nose—the "butterfly" rash), ears, chin, V region of the neck and chest, upper back, and extensor surfaces of the arms. Worsening of this rash often accompanies flare of systemic disease. Subacute cutaneous lupus erythematosus (SCLE) consists of scaly red patches similar to psoriasis, or circular flat red-rimmed lesions. Patients with these manifestations are exquisitely photosensitive; most have antibodies to Ro (SS-A). Other SLE rashes include recurring urticaria, lichen planus-like dermatitis, bullae, and panniculitis ("lupus profundus"). Rashes can be minor or severe; they may be the major disease manifestation. Small ulcerations on the oral or nasal mucosa are common in SLE; the lesions resemble aphthous ulcers and may or may not be painful.

RENAL MANIFESTATIONS

Nephritis is usually the most serious manifestation of SLE, particularly because nephritis and infection are the leading causes of mortality in the first decade of disease. Because nephritis is asymptomatic in most lupus patients, urinalysis should be ordered in any person suspected of having SLE. The classification of lupus nephritis is primarily histologic (see "Pathology," above, and Table 349-2). Renal biopsy is recommended for every SLE patient with any clinical evidence of nephritis; results are used to plan current and near-future therapies. Patients with dangerous proliferative forms of glomerular damage (ISN III and IV) usually have microscopic hematuria and proteinuria (>500 mg per 24 h); approximately one-half develop nephrotic syndrome, and most develop hypertension. If diffuse proliferative glomerulonephritis (DPGN) is inadequately treated, virtually all patients develop ESRD within 2 years of diagnosis. Therefore, aggressive immunosuppression is indicated (usually systemic glucocorticoids plus another immunosuppressive drug), unless damage is irreversible (Fig. 349-2, Table 349-5). African Americans are more likely to develop ESRD than are whites, even with the most current therapies. Overall in the United States, ~20% of individuals with lupus DPGN die or develop ESRD within 10 years of diagnosis. Such individuals require aggressive control of SLE and of the complications of renal disease and of therapy. Approximately 20% of SLE patients with proteinuria (usually nephrotic) have membranous glomerular changes without proliferative changes on renal biopsy. Their outcome is better than for those with DPGN, but patients with class V and nephrotic range proteinuria should be treated in the same way as those with classes III or IV proliferative disease. Lupus nephritis tends to be an ongoing disease, with flares requiring re-treatment or increased treatment over many years. For most people with lupus nephritis, accelerated atherosclerosis becomes important after several years of disease; attention must be given to control of systemic inflammation, blood pressure, hyperlipidemia, and hyperglycemia.

NERVOUS SYSTEM MANIFESTATIONS

There are many central nervous system (CNS) and peripheral nervous system manifestations of SLE; in some patients, these are the major cause of morbidity and mortality. It is useful to approach this diagnostically by asking first whether the symptoms result from SLE or another condition (such as infection in immunosuppressed individuals or side effects of therapies). If symptoms are related to SLE, it should be determined whether they are caused by a diffuse process (requiring immunosuppression) or vascular occlusive disease (requiring anticoagulation). The most common manifestation of diffuse CNS lupus is

TABLE 349-4 Clinical Manifestations of SLE and Prevalence over the Entire Course of Disease[a]

MANIFESTATION	PREVALENCE, %
Systemic: Fatigue, malaise, fever, anorexia, weight loss	95
Musculoskeletal	95
Arthralgias/myalgias	95
Nonerosive polyarthritis	60
Hand deformities	10
Myopathy/myositis	25/5
Ischemic necrosis of bone	15
Cutaneous	80
Photosensitivity	70
Malar rash	50
Oral ulcers	40
Alopecia	40
Discoid rash	20
Vasculitis rash	20
Other (e.g., urticaria, subacute cutaneous lupus)	15
Hematologic	85
Anemia (chronic disease)	70
Leukopenia (<4000/μL)	65
Lymphopenia (<1500/μL)	50
Thrombocytopenia (<100,000/μL)	15
Lymphadenopathy	15
Splenomegaly	15
Hemolytic anemia	10
Neurologic	60
Cognitive disorder	50
Mood disorder	40
Depression	25
Headache	25
Seizures	20
Mono-, polyneuropathy	15
Stroke, TIA	10
Acute confusional state or movement disorder	2–5
Aseptic meningitis, myelopathy	<1
Cardiopulmonary	60
Pleurisy, pericarditis, effusions	30–50
Myocarditis, endocarditis	10
Lupus pneumonitis	10
Coronary artery disease	10
Interstitial fibrosis	5
Pulmonary hypertension, ARDS, hemorrhage	<5
Shrinking lung syndrome	<5
Renal	30–50
Proteinuria ≥500 mg/24 h, cellular casts	30–60
Nephrotic syndrome	25
End-stage renal disease	5–10
Gastrointestinal	40
Nonspecific (nausea, mild pain, diarrhea)	30
Abnormal liver enzymes	40
Vasculitis	5
Thrombosis	15
Venous	10
Arterial	5
Ocular	15
Sicca syndrome	15
Conjunctivitis, episcleritis	10
Vasculitis	5

[a]Numbers indicate percentage of patients who have the manifestation at some time during the course of illness.

Abbreviations: ARDS, acute respiratory distress syndrome; SLE, Systemic Lupus Erythematosus; TIA, transient ischemic attack.

cognitive dysfunction, including difficulties with memory and reasoning. Headaches are also common. When excruciating, they often indicate SLE flare; when milder, they are difficult to distinguish from migraine or tension headaches. Seizures of any type may be caused by lupus; treatment often requires both antiseizure and immunosuppressive therapies. Psychosis can be the dominant manifestation of SLE; it must be distinguished from glucocorticoid-induced psychosis. The latter usually occurs in the first weeks of glucocorticoid therapy, at daily doses of ≥40 mg of prednisone or equivalent; psychosis resolves over several days after glucocorticoids are decreased or stopped. Myelopathy is not rare and is often disabling; rapid initiation of immunosuppressive therapy starting with high-dose glucocorticoids is standard of care.

VASCULAR OCCLUSIONS INCLUDING STROKE AND MYOCARDIAL INFARCTIONS

The prevalence of transient ischemic attacks, strokes, and myocardial infarctions is increased in patients with SLE. These vascular events are increased in, but not exclusive to, SLE patients with antibodies to phospholipids (antiphospholipid antibodies), which are associated with hypercoagulability and acute thrombotic events **(Chap. 350)**. Ischemia in the brain can be caused by focal occlusion (either noninflammatory or associated with vasculitis) or by embolization from carotid artery plaque or from fibrinous vegetations of Libman-Sacks endocarditis. Appropriate tests for antiphospholipid antibodies (see below) and for sources of emboli should be ordered in such patients to estimate the need for, intensity of, and duration of anti-inflammatory and/or anticoagulant therapies. When it is most likely that a cerebral event results from clotting, long-term anticoagulation is the therapy of choice. Two processes can occur at once—vasculitis plus bland vascular occlusions—in which case it is appropriate to treat with anticoagulation plus immunosuppression.

In SLE, myocardial infarctions are primarily manifestations of accelerated atherosclerosis. The increased risk for vascular events is three- to tenfold overall, and is highest in women aged <49 years. Characteristics associated with increased risk for atherosclerosis include male gender, older age, hypertension, dyslipidemia, diabetes, dysfunctional proinflammatory high-density lipoproteins, repeated high scores for disease activity, high cumulative or daily doses of glucocorticoids, and high serum levels of homocysteine and leptin. Statin therapies reduce levels of low-density lipoproteins (LDL) in SLE patients; significant reduction of cardiac events by statins has been shown in SLE patients with renal transplants and recently in an epidemiologic study of large number of patients in Taiwan.

PULMONARY MANIFESTATIONS

The most common pulmonary manifestation of SLE is pleuritis with or without pleural effusion. This manifestation, when mild, may respond to treatment with nonsteroidal anti-inflammatory drugs (NSAIDs); when more severe, patients require a brief course of glucocorticoid therapy. Pulmonary infiltrates also occur as a manifestation of active SLE and are difficult to distinguish from infection on imaging studies. Life-threatening pulmonary manifestations include interstitial inflammation leading to fibrosis, shrinking lung syndrome, and intraalveolar hemorrhage; all of these probably require early aggressive immunosuppressive therapy as well as supportive care. Pulmonary arterial hypertension occurs in a small proportion of SLE patients and should be treated in the same way as idiopathic pulmonary hypertension.

CARDIAC MANIFESTATIONS

Pericarditis is the most frequent cardiac manifestation; it usually responds to anti-inflammatory therapy and infrequently leads to tamponade. More serious cardiac manifestations are myocarditis and fibrinous endocarditis of Libman-Sacks. The endocardial involvement can lead to valvular insufficiencies, most commonly of the mitral or aortic valves, or to embolic events. It has not been proven that glucocorticoid or other immunosuppressive therapies lead to improvement of lupus myocarditis or endocarditis, but it is usual practice to administer a trial of high-dose steroids along with appropriate supportive

TABLE 349-5 Medications for the Management of SLE

MEDICATION	DOSE RANGE	DRUG INTERACTIONS	SERIOUS OR COMMON ADVERSE EFFECTS
NSAIDs, salicylates (Ecotrin[a] and St. Joseph's aspirin[a] approved by FDA for use in SLE)	Doses toward upper limit of recommended range usually required	A2R/ACE inhibitors, glucocorticoids, fluconazole, methotrexate, thiazides	NSAIDs: Higher incidence of aseptic meningitis, elevated liver enzymes, decreased renal function, vasculitis of skin; entire class, especially COX-2-specific inhibitors, may increase risk for myocardial infarction Salicylates: ototoxicity, tinnitus Both: GI events and symptoms, allergic reactions, dermatitis, dizziness, acute renal failure, edema, hypertension
Topical glucocorticoids	Mid potency for face; mid to high potency for other areas	None known	Atrophy of skin, contact dermatitis, folliculitis, hypopigmentation, infection
Topical sunscreens	SPF 15 at least; 30+ preferred	None known	Contact dermatitis
Hydroxychloroquine[a] (quinacrine can be added or substituted)	200–400 mg qd (100 mg qd); do not exceed 6.5 mg/kg dry weight	None known	Retinal damage, agranulocytosis, aplastic anemia, ataxia, cardiomyopathy, dizziness, myopathy, ototoxicity, peripheral neuropathy, pigmentation of skin, seizures, thrombocytopenia; Quinacrine usually causes diffuse yellow skin coloration
DHEA (dehydroepiandrosterone)	200 mg qd	Unclear	Acne, menstrual irregularities, high serum levels of testosterone
Methotrexate (for dermatitis, arthritis)	10–25 mg once a week, PO or SC, with folic acid; decrease dose if CrCl <60 mL/min	Acitretin, leflunomide, NSAIDs and salicylates, penicillins, probenecid, sulfonamides, trimethoprim	Anemia, bone marrow suppression, leukopenia, thrombocytopenia, hepatotoxicity, nephrotoxicity, infections, neurotoxicity, pulmonary fibrosis, pneumonitis, severe dermatitis, seizures, pseudolymphoma
Glucocorticoids, oral[a] (several specific brands are approved by FDA for use in SLE)	Prednisone, prednisolone: 0.5–1 mg/kg per day for severe SLE 0.07–0.3 mg/kg per day or qod for milder disease	A2R/ACE antagonists, antiarrhythmics class III, cyclosporine, NSAIDs and salicylates, phenothiazines, phenytoins, quinolones, rifampin, risperidone, thiazides, sulfonylureas, warfarin	Infection, VZV infection, hypertension, hyperglycemia, hypokalemia, acne, allergic reactions, anxiety, aseptic necrosis of bone, cushingoid changes, CHF, fragile skin, insomnia, menstrual irregularities, mood swings, osteoporosis, psychosis
Methylprednisolone sodium succinate, IV[a] (FDA approved for lupus nephritis)	For severe disease, 0.5-1 g IV qd × 3 days	As for oral glucocorticoids	As for oral glucocorticoids (if used repeatedly); anaphylaxis
Cyclophosphamide[b] IV	Low dose (for whites of northern European backgrounds): 500 mg every 2 weeks for 6 doses, then begin maintenance with MMF or AZA. High dose: 7–25 mg/kg q month × 6; consider mesna administration with dose	Allopurinol, bone marrow suppressants, colony-stimulating factors, doxorubicin, rituximab, succinylcholine, zidovudine	Infection, VZV infection, bone marrow suppression, leukopenia, anemia, thrombocytopenia, hemorrhagic cystitis (less with IV), carcinoma of the bladder, alopecia, nausea, diarrhea, malaise, malignancy, ovarian and testicular failure. Ovarian failure is probably not a problem with low dose.
Oral	1.5–3 mg/kg per day; decrease dose for CrCl <25 mL/min		
Mycophenolate mofetil (MMF)[b] or mycophenolic acid (MPA)	MMF: 2–3 g/d PO total given bid for induction therapy, 1–2 g/d total given bid for maintenance therapy; max 1 g bid if CrCl <25 mL/min. Begin with low dose and increase every 1–2 weeks to minimize GI side effects. Start treatment at 0.5 g bid. MPA: 360–1080 mg bid; caution if CrCl <25 mL/min	Acyclovir, antacids, azathioprine, bile acid-binding resins, ganciclovir, iron, salts, probenecid, oral contraceptives	Infection, leukopenia, anemia, thrombocytopenia, lymphoma, lymphoproliferative disorders, malignancy, alopecia, cough, diarrhea, fever, GI symptoms, headache, hypertension, hypercholesterolemia, hypokalemia, insomnia, peripheral edema, elevated liver enzymes, tremor, rash. Limited date suggests Asians should begin treatment with doses not exceeding 2 g daily to reduce adverse events.
Azathioprine (AZA)[b]	2–3 mg/kg per day PO for induction; 1–2 mg/kg per day for maintenance; decrease frequency of dose if CrCl <50 mL/min	ACE inhibitors, allopurinol, bone marrow suppressants, interferons, mycophenolate mofetil, rituximab, warfarin, zidovudine	Infection, VZV infection, bone marrow suppression, leukopenia, anemia, thrombocytopenia, pancreatitis, hepatotoxicity, malignancy, alopecia, fever, flulike illness, GI symptoms
Belimumab	10 mg/kg IV wks 0, 2, and 4, then monthly OR subcutaneous 200 mg each week	IVIg	Infusion reactions, allergy, infections. Headache and diffuse body aching.
Rituximab (for patients resistant to above therapies)	375 mg/m^2 q wk × 4 or 1 g q 2 wks × 2	IVIg	Infection (including PML), infusion reactions, headache, arrhythmias, allergic responses
Tacrolimus	Trough blood level should not exceed 5.5 ng/mL to minimize toxicity. Begin dose at 2 mg bid		Infection, nephrotoxicity, neural toxicity

[a]Indicates medication is approved for use in SLE by the U.S. Food and Drug Administration. [b]Indicates the medication has been used with glucocorticoids in the trials showing efficacy.

Abbreviations: A2R, angiotensin II receptor; ACE, angiotensin-converting enzyme; CHF, congestive heart failure; CrCl, creatinine clearance; FDA, U.S. Food and Drug Administration; GI, gastrointestinal; IVIg, intravenous immunoglobulin; NSAIDs, nonsteroidal anti-inflammatory drugs; PML, progressive multifocal leukoencephalopathy; SLE, systemic lupus erythematosus; SPF, sun protection factor; VZV, varicella-zoster virus.

therapy for heart failure, arrhythmia, or embolic events. As discussed above, patients with SLE are at increased risk for myocardial infarction, usually due to accelerated atherosclerosis, which probably results from immune attack, chronic inflammation, and/or chronic oxidative damage to arteries.

HEMATOLOGIC MANIFESTATIONS

The most frequent hematologic manifestation of SLE is anemia, usually normochromic normocytic, reflecting chronic illness. Hemolysis can be rapid in onset and severe, requiring high-dose glucocorticoid therapy, which is effective in most patients. Leukopenia is also common and almost always consists of lymphopenia, not granulocytopenia; lymphopenia rarely predisposes to infections and by itself usually does not require therapy. Thrombocytopenia may be a recurring problem. If platelet counts are >40,000/μL and abnormal bleeding is absent, therapy may not be required. High-dose glucocorticoid therapy (e.g., 1 mg/kg per day of prednisone or equivalent) is usually effective for the first few episodes of severe thrombocytopenia. Recurring or prolonged hemolytic anemia or thrombocytopenia, or disease requiring an unacceptably high dose of daily glucocorticoids, should be treated with additional strategies such as rituximab, platelet growth factors, and/or splenectomy (see "Management of Systemic Lupus Erythematosus" below).

GASTROINTESTINAL MANIFESTATIONS

Nausea, sometimes with vomiting, and diarrhea can be manifestations of an SLE flare, as can diffuse abdominal pain probably caused by autoimmune peritonitis and/or intestinal vasculitis. Increases in serum aspartate aminotransferase (AST) and alanine aminotransferase (ALT) are common when SLE is active. These manifestations usually improve promptly during systemic glucocorticoid therapy. Vasculitis involving the intestine may be life-threatening; perforations, ischemia, bleeding, and sepsis are frequent complications. Aggressive immunosuppressive therapy with high-dose glucocorticoids is recommended for short-term control; evidence of recurrence is an indication for additional therapies.

OCULAR MANIFESTATIONS

Sicca syndrome (Sjögren's syndrome; **Chap. 354**) and nonspecific conjunctivitis are common in SLE and rarely threaten vision. In contrast, retinal vasculitis and optic neuritis are serious manifestations: blindness can develop over days to weeks. Aggressive immunosuppression is recommended, although there are no controlled trials to prove effectiveness. Complications of systemic and intraorbital glucocorticoid therapy include cataracts (common) and glaucoma.

LABORATORY TESTS

Laboratory tests serve (1) to establish or rule out the diagnosis; (2) to follow the course of disease, particularly to suggest that a flare is occurring or organ damage is developing; and (3) to identify adverse effects of therapies.

TESTS FOR AUTOANTIBODIES (TABLES 349-1 AND 349-3)

Diagnostically, the most important autoantibodies to detect are ANA because the test is positive in >95% of patients, usually at the onset of symptoms. A few patients develop ANA within 1 year of symptom onset; repeated testing may thus be useful. ANA tests using immunofluorescent methods are more reliable than enzyme-linked immunosorbent assays (ELISAs) and/or bead assays, which have less specificity. ANA-negative lupus exists but is rare in adults and is usually associated with other autoantibodies (anti-Ro or anti-DNA). High-titer IgG antibodies to double-stranded DNA (dsDNA) (but not to single-stranded DNA) are specific for SLE. ELISA and immunofluorescent reactions of sera with the dsDNA in the flagellate *Crithidia luciliae* have ~60% sensitivity for SLE. Titers of anti-dsDNA vary over time. In some patients, increases in quantities of anti-dsDNA herald a flare, particularly of nephritis or vasculitis, especially when associated with declining levels of C3 or C4 complement. Antibodies to Sm are also specific for SLE and assist in diagnosis; anti-Sm antibodies do not usually correlate with disease activity or clinical manifestations. Antiphospholipid antibodies are not specific for SLE, but their presence fulfills one classification criterion, and they identify patients at increased risk for venous or arterial clotting, thrombocytopenia, and fetal loss. There are three widely accepted tests that measure different antibodies (anticardiolipin, anti-β_2-glycoprotein, and the lupus anticoagulant). ELISA is used for anticardiolipin and anti-β_2-glycoprotein (both internationally standardized with good reproducibility); a sensitive phospholipid-based activated prothrombin time such as the dilute Russell venom viper test is used to identify the lupus anticoagulant. The higher the titers of IgG anticardiolipin (>40 IU is considered high), and the greater the number of different antiphospholipid antibodies that are detected, the greater is the risk for a clinical episode of clotting. Quantities of antiphospholipid antibodies may vary markedly over time; repeated testing is justified if clinical manifestations of the APS appear **(Chap. 350)**. To classify a patient as having APS, with or without SLE, by international criteria requires the presence of one or more clotting episodes and/or repeated fetal losses plus at least two positive tests for antiphospholipid antibodies, at least 12 weeks apart; however, many patients with APS do not meet these stringent criteria, which are intended for inclusion of patients into studies.

An additional autoantibody test with predictive value (not used for diagnosis) detects anti-Ro/SS-A, which indicates increased risk for neonatal lupus, sicca syndrome, and SCLE. Women with child-bearing potential and SLE should be screened for antiphospholipid antibodies and anti-Ro, because both antibodies have the potential to cause fetal harm.

STANDARD TESTS FOR DIAGNOSIS

Screening tests for complete blood count, platelet count, and urinalysis may detect abnormalities that contribute to the diagnosis and influence management decisions.

TESTS FOR FOLLOWING DISEASE COURSE

It is useful to follow tests that indicate the status of organ involvement known to be present during SLE flares. These might include urinalysis for hematuria and proteinuria, hemoglobin levels, platelet counts, and serum levels of creatinine or albumin. There is great interest in identification of additional markers of disease activity. Candidates include levels of anti-DNA and anti-C1q antibodies, several components of complement (C3 is most widely available), activated complement products (an assay is commercially available that measures binding to the C4d receptor on erythrocytes and B cells), IFN-inducible gene expression in peripheral blood cells, serum levels of BLyS (B lymphocyte stimulator, also called BAFF), and urinary levels of TNF-like weak inducer of apoptosis (TWEAK), neutrophil gelatinase-associated lipocalin (NGAL), or monocyte chemotactic protein 1 (MCP-1). None is uniformly agreed upon as a reliable indicator of flare or of response to therapeutic interventions. It is likely that a panel of multiple proteins and nuclear products (and possibly levels of selected miRNAs and methylation profiles of DNA) will be developed to predict both impending flare and response to recently instituted therapies. Increased quantities of plasma cells, and increased expression of their gene signatures in whole blood, are associated with active disease and flares, but measurements are not commercially available. For now, the physician should determine for each patient whether certain available laboratory test changes predict flare (falling complement, rising anti-DNA, increased proteinuria, worsening anemia, etc.). If so, altering therapy in response to these changes may be advisable (30 mg of prednisone daily for 2 weeks has been shown to prevent flares in patients with rising anti-DNA plus falling complement). In addition, given the increased prevalence of atherosclerosis in SLE, it is advisable to follow the recommendations of the National Cholesterol Education Program for testing and treatment, including scoring of SLE as an independent risk factor, similar to diabetes mellitus.

MANAGEMENT OF SYSTEMIC LUPUS ERYTHEMATOSUS

There is no cure for SLE, and complete sustained remissions are rare. There is an international effort to encourage practitioners and patients to aim for low-level disease activity (mild symptoms on the lowest

possible doses of medications) which can be achieved for at least a year in 30–50% of SLE patients. Therefore, the physician should plan to induce remissions of acute flares and then maintain improvements with strategies that suppress symptoms to an acceptable level and prevent organ damage. Therapeutic choices depend on (1) whether disease manifestations are life-threatening or likely to cause organ damage, justifying aggressive therapies; (2) whether manifestations are potentially reversible; and (3) the best approaches to preventing complications of disease and its treatments. Therapies, doses, and adverse effects are listed in Table 349-5.

CONSERVATIVE THERAPIES FOR MANAGEMENT OF NON-LIFE-THREATENING DISEASE

Among patients with fatigue, pain, and autoantibodies indicative of SLE, but without major organ involvement, management can be directed to suppression of symptoms. Analgesics and antimalarials are mainstays. NSAIDs are useful analgesics/anti-inflammatories, particularly for arthritis/arthralgias. However, two major issues indicate caution in using NSAIDs. First, SLE patients compared with the general population are at increased risk for NSAID-induced aseptic meningitis, elevated serum transaminases, hypertension, and renal dysfunction. Second, all NSAIDs, particularly those that inhibit cyclooxygenase-2 specifically, may increase risk for myocardial infarction. Acetaminophen to control pain may be a good strategy, but NSAIDs are more effective in some patients. The relative hazards of NSAIDs compared with low-dose glucocorticoid therapy have not been established. Antimalarials (hydroxychloroquine, chloroquine, and quinacrine) often reduce dermatitis, arthritis, and fatigue. A randomized, placebo-controlled, prospective trial has shown that withdrawal of hydroxychloroquine results in increased numbers of disease flares; hydroxychloroquine also reduces accrual of tissue damage, including renal damage, over time. Some experts recommend a hydroxychloroquine blood level of ≥750 ng/mL to optimize responses in active SLE; after achieving response doses should be reduced. Because of potential retinal toxicity (occurring in 6% of patients after cumulative doses of 1000 g, ~5 years of continuing therapy), patients receiving antimalarials should undergo ophthalmologic examinations annually. A placebo-controlled prospective trial suggests that administration of dehydroepiandrosterone may reduce disease activity. If quality of life is inadequate despite these conservative measures, treatment with low doses of systemic glucocorticoids may be necessary. Belimumab is effective for 50% of patients with fatigue, rash, and/or the arthritis of SLE; it is expensive and should be considered after other approaches fail or are not tolerated. SLE patients most likely to respond to belimumab have robust clinical activity (a Systemic Lupus Erythematosus Disease Activity Index [SLEDAI] score of ≥10), positive anti-DNA, and low serum complement. SLEDAI is a widely used measure of SLE disease activity; scores >3 reflect clinically active disease. Lupus dermatitis should be managed with topical sunscreens, antimalarials, topical glucocorticoids and/or tacrolimus, and if severe or unresponsive, systemic glucocorticoids with or without mycophenolate mofetil, azathioprine, or belimumab.

LIFE-THREATENING SLE: PROLIFERATIVE FORMS OF LUPUS NEPHRITIS

Guidelines for management of lupus nephritis have been published recently by the American College of Rheumatology and the European League Against Rheumatism (encompassed and referenced in Fig. 349-2 and Table 349-5). The mainstay of treatment for any inflammatory life-threatening or organ-threatening manifestations of SLE is systemic glucocorticoids (0.5–1 mg/kg per day PO or 500–1000 mg of methylprednisolone sodium succinate IV daily for 3 days followed by 0.5–1 mg/kg of daily prednisone or equivalent). Evidence that glucocorticoid therapy is life-saving comes from retrospective studies from the predialysis era; survival was significantly better in people with DPGN treated with high-dose daily glucocorticoids (40–60 mg of prednisone daily for 4–6 months) versus lower doses. Currently, high doses are recommended for much shorter periods; recent trials of interventions for severe SLE use 4–6 weeks of 0.5–1 mg/kg per day of prednisone or equivalent. Thereafter, doses are tapered as rapidly as the clinical situation permits, usually to a maintenance dose ranging from 5 to 10 mg of prednisone or equivalent per day. Most patients with an episode of severe SLE require many years of maintenance therapy with low-dose glucocorticoids, which can be increased to prevent or treat disease flares. Frequent attempts to gradually reduce the glucocorticoid requirement are recommended because virtually everyone develops important adverse effects (Table 349-5). High-quality clinical studies regarding initiating therapy for severe, active SLE with IV pulses of high-dose glucocorticoids are not available. Most recent clinical trials in lupus nephritis have initiated therapy with high-dose IV glucocorticoid pulses (500–1000 mg daily for 3–5 days). This approach must be tempered by safety considerations, such as the presence of conditions adversely affected by glucocorticoids (e.g., infection, hyperglycemia, hypertension, osteoporosis). A current clinical trial is evaluating mycophenolate mofetil plus rituximab without maintenance daily glucocorticoids to treat lupus nephritis: if results are positive, the paradigm for short-term and long-term management of SLE is likely to change.

Cytotoxic/immunosuppressive agents added to glucocorticoids are recommended to treat serious SLE. Almost all prospective controlled trials in SLE involving cytotoxic agents have been conducted in combination with glucocorticoids in patients with lupus nephritis. Therefore, the following recommendations apply to treatment of nephritis. Either cyclophosphamide (an alkylating agent) or mycophenolate mofetil (a relatively lymphocyte-specific inhibitor of inosine monophosphatase and therefore of purine synthesis) is an acceptable choice for induction of improvement in severely ill patients; azathioprine (a purine analogue and cycle-specific antimetabolite) may be effective but is associated with more flares. In patients whose renal biopsies show ISN grade III or IV disease, early treatment with combinations of glucocorticoids and cyclophosphamide reduces progression to ESRD and death. Shorter-term studies with glucocorticoids plus mycophenolate mofetil (prospective randomized trials of 6 months, follow-up studies of 5 years) show that this regimen is similar to cyclophosphamide in achieving improvement. Comparisons are complicated by effects of race, since higher proportions of African Americans (and other non-Asian, nonwhite races) respond to mycophenolate than to cyclophosphamide, whereas similar proportions of whites and Asians respond to each drug. Regarding toxicity, diarrhea is more common with mycophenolate mofetil; amenorrhea, leukopenia, and nausea are more common with high dose cyclophosphamide. Importantly, rates of severe infections and death are similar in meta-analyses. Two different regimens of IV cyclophosphamide are available. For white patients with northern European backgrounds, low doses of cyclophosphamide (500 mg every 2 weeks for six total doses, followed by azathioprine or mycophenolate maintenance) are as effective as standard high doses, with less toxicity. Ten-year follow-up has shown no differences between the high-dose and low-dose groups (death or ESRD in 9–20% of patients in each group). It is not clear whether the data apply to U.S. populations, especially African Americans and Latinas. High-dose cyclophosphamide (500–1000 mg/m^2 body surface area given monthly IV for 6 months, followed by azathioprine or mycophenolate maintenance) is an acceptable approach for patients with severe nephritis (e.g., multiple cellular crescents and/or fibrinoid necrosis on renal biopsy, or rapidly progressive glomerulonephritis). Cyclophosphamide and mycophenolate responses begin 3–16 weeks after treatment is initiated, whereas glucocorticoid responses may begin within 24 h.

For maintenance therapy, mycophenolate and azathioprine probably are similar in efficacy and toxicity; both are safer than cyclophosphamide. In a recently published multicenter study, mycophenolate was superior to azathioprine in maintaining renal function and survival in patients who responded to induction therapy with either cyclophosphamide or mycophenolate. The incidence of ovarian failure, a common effect of high-dose cyclophosphamide therapy (but probably not of low-dose therapy), can be reduced by treatment with a gonadotropin-releasing hormone agonist (e.g., leuprolide 3.75 mg intramuscularly) prior to each monthly cyclophosphamide dose. Patients with high serum creatinine levels (e.g., ≥265 μmol/L [≥3.0 mg/dL]) many months in duration and high chronicity scores on renal biopsy are not

likely to respond to any of these therapies. In general, it may be better to induce improvement in an African-American or Hispanic patient with proliferative glomerulonephritis with mycophenolate mofetil (2–3 g daily) rather than cyclophosphamide, with the option to switch if no evidence of response is detectable after 3–6 months of treatment. For whites and Asians, induction with either mycophenolate mofetil or cyclophosphamide is acceptable. Cyclophosphamide may be discontinued when it is clear that a patient is improving. The number of SLE flares is reduced by maintenance therapy with mycophenolate mofetil (1.5–2 g daily) or azathioprine (1–2.5 mg/kg per day). Both cyclophosphamide and mycophenolate mofetil are potentially teratogenic; patients should be off either medication for at least 3 months before attempting to conceive. Azathioprine can be used if necessary to control active SLE in patients who are pregnant. If azathioprine is used either for induction or maintenance therapy, patients may be prescreened for homozygous deficiency of the TMPT enzyme (which is required to metabolize the 6-mercaptopurine product of azathioprine) because they are at higher risk for bone marrow suppression.

Good improvement occurs in ~80% of lupus nephritis patients receiving either cyclophosphamide or mycophenolate at 1–2 years of follow-up. However, in some studies, at least 50% of these individuals have flares of nephritis over the next 5 years, and re-treatment is required; such individuals are more likely to progress to ESRD. Long-term outcome of lupus nephritis to most interventions is better in whites than in African Americans. Methotrexate (a folinic acid antagonist) may have a role in the treatment of arthritis and dermatitis but probably not in nephritis or other life-threatening disease. Small controlled trials (in Asia) of leflunomide, a relatively lymphocyte-specific pyrimidine antagonist licensed for use in rheumatoid arthritis, have suggested it can suppress disease activity in some SLE patients. Cyclosporine and tacrolimus, which inhibit calcium flux and therefore production of IL-2 and T lymphocyte functions, have not been studied in prospective controlled trials in SLE in the United States; several studies in Asia have shown they are effective in lupus nephritis. A recent trial in China showed that a combination of low dose mycophenolate mofetil (one gram daily) plus tacrolimus (4 mg daily) plus prednisone (pulse followed by 0.6 mg/kg/d) had a better response rate than high dose intravenous cyclophosphamide. Because calcineurin blockers have potential nephrotoxicity but little bone marrow toxicity, the author uses them for periods of a few months in patients with steroid-resistant cytopenias of SLE, in steroid-resistant patients who have developed bone marrow suppression from standard cytotoxic agents, or in patients with active SLE in spite of treatment with mycophenolate or cyclosphosphamide.

Most patients with SLE of any type should be treated with hydroxychloroquine since it prevents damage in skin and kidney and reduces overall damage scores. Patients with proteinuria > 500 mg daily should receive ACE inhibitors or ARBs, as they reduce the chance for ESRD.

Use of biologics directed against B cells for active SLE is under intense study. Use of anti-CD20 (rituximab), particularly in patients with SLE who are resistant to the more standard combination therapies discussed above, is controversial. Several open trials have shown efficacy in a majority of such patients, both for nephritis and for extrarenal lupus. However, recent prospective placebo-controlled randomized trials, one in renal and one in nonrenal SLE, did not show a difference between anti-CD20 and placebo when added to standard combination therapies. Belimumab, which is approved by the FDA for use in SLE without active renal disease (indication is serologically positive SLE which has failed standard treatments), is in clinical trials for active lupus nephritis. Drugs that kill plasma cells, used in multiple myeloma, are being studied in SLE, as are molecules and antibodies that prevent activation of B cells and/or T cells, such as Jak/Stat inhibitors.

SPECIAL CONDITIONS IN SLE THAT MAY REQUIRE ADDITIONAL OR DIFFERENT THERAPIES

Crescentic Lupus Nephritis The presence of cellular or fibrotic crescents in glomeruli with proliferative glomerulonephritis indicates a worse prognosis than in patients without this feature. There are no large prospective multinational controlled trials showing efficacy of cyclophosphamide, mycophenolate, cyclosporine, or tacrolimus in such cases. Most authorities recommend high-dose cyclophosphamide as the induction therapy of choice; there is some evidence that high dose mycophenolate mofetil is equally effective.

Membranous Lupus Nephritis Most SLE patients with membranous (INS-V) nephritis also have proliferative changes and should be treated for proliferative disease. However, some have pure membranous changes. Treatment for this group is less well defined. Some authorities do not recommend immunosuppression unless proteinuria is in the nephrotic range (although treatment with angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers is recommended). In those patients, recent prospective controlled trials suggest that alternate-day glucocorticoids plus cyclophosphamide or mycophenolate mofetil or cyclosporine are all effective in the majority of patients in reducing proteinuria. It is more controversial whether they preserve renal function over the long term.

Pregnancy and Lupus Fertility rates for men and women with SLE are probably normal. However, rate of fetal loss is increased (approximately two- to threefold) in women with SLE. Fetal demise is higher in mothers with high disease activity, antiphospholipid antibodies (especially the lupus anticoagulant), hypertension, and/or active nephritis. Suppression of disease activity can be achieved by administration of systemic glucocorticoids. A placental enzyme, 11-β-dehydrogenase 2, deactivates glucocorticoids; it is more effective in deactivating prednisone and prednisolone than the fluorinated glucocorticoids dexamethasone and betamethasone. Glucocorticoids are listed by the FDA as pregnancy category A (no evidence of teratogenicity in human studies); cyclosporine, tacrolimus, and rituximab are listed as category C (may be teratogenic in animals but no good evidence in humans); azathioprine, hydroxychloroquine, mycophenolate mofetil, and cyclophosphamide are category D (there is evidence of teratogenicity in humans, but benefits might outweigh risks in certain situations); and methotrexate is category X (risks outweigh benefits). Therefore, active SLE in pregnant women should be controlled with hydroxychloroquine and, if necessary, prednisone/prednisolone at the lowest effective doses for the shortest time required. Azathioprine may be added if these treatments do not suppress disease activity. Adverse effects of prenatal glucocorticoid exposure (primarily betamethasone, which is not recommended) on offspring may include low birth weight, developmental abnormalities in the CNS, and predilection toward adult metabolic syndrome. It is likely that each of these glucocorticoids and immunosuppressive medications gets into breast milk, at least in low levels; patients should consider not breastfeeding if they need therapy for SLE. In SLE patients with antiphospholipid antibodies and prior fetal losses, treatment with heparin (usually low-molecular-weight) plus low-dose aspirin has been shown in prospective controlled trials to increase significantly the proportion of live births. Aspirin alone may be used, although most consider it less effective than heparin-plus-aspirin. Warfarin is teratogenic. Studies with oral thrombin and Factor Xa inhibitors are in progress for APSs; however, their role in preventing fetal loss is not established. An additional potential problem for the fetus is the presence of antibodies to Ro, sometimes associated with neonatal lupus consisting of rash and congenital heart block with or without cardiomyopathy. The cardiac manifestations can be life-threatening; therefore the presence of anti-Ro requires vigilant monitoring of fetal heart rates with prompt intervention (delivery if possible) if distress occurs. Recent evidence shows that hydroxychloroquine treatment of an anti-Ro-positive mother whose infant develops congenital heart block significantly reduces the chance that subsequent fetuses will develop heart block. There is some evidence that dexamethasone treatment of a mother in whom first- or second-degree heart block is detected in utero sometimes prevents progression of heart block. Women with SLE usually tolerate pregnancy without disease flares. However, a small proportion develops severe flares requiring aggressive glucocorticoid therapy or early delivery.

Lupus and Antiphospholipid Syndrome Patients with SLE who have venous or arterial clotting and/or repeated fetal losses and

at least two positive tests for antiphospholipid antibodies have APS and should be managed with long-term anticoagulation (Chap. 350). With warfarin, a target international normalized ratio (INR) of 2.0–2.5 is recommended for patients with one episode of venous clotting; an INR of 3.0–3.5 is recommended for patients with recurring clots or arterial clotting, particularly in the CNS. Recommendations are based on both retrospective and prospective studies of posttreatment clotting events and adverse effects from anticoagulation. Thrombin and Factor Xa inhibitors are under study.

Microvascular Thrombotic Crisis (Thrombotic Thrombocytopenic Purpura, Hemolytic-Uremic Syndrome) This syndrome of hemolysis, thrombocytopenia, and microvascular thrombosis in kidneys, brain, and other tissues carries a high mortality rate and occurs most commonly in young individuals with lupus nephritis. The most useful laboratory tests are identification of schistocytes on peripheral blood smears, elevated serum levels of lactate dehydrogenase, and antibodies to ADAMS13. Plasma exchange or extensive plasmapheresis is usually life-saving; most authorities recommend concomitant glucocorticoid therapy; there is no evidence that cytotoxic drugs are effective.

Lupus Dermatitis Patients with any form of lupus dermatitis should minimize exposure to ultraviolet light, using appropriate clothing and sunscreens with a sun protection factor of at least 30. Topical glucocorticoids and antimalarials (such as hydroxychloroquine) are effective in reducing lesion severity in most patients and are relatively safe. Systemic treatment with retinoic acid is a useful strategy in patients with inadequate improvement on topical glucocorticoids and antimalarials; adverse effects are potentially severe (particularly fetal abnormalities), and there are stringent reporting requirements for its use in the United States. Extensive, pruritic, bullous, or ulcerating dermatitides usually improve promptly after institution of systemic glucocorticoids; tapering may be accompanied by flare of lesions, thus necessitating use of a second medication such as hydroxychloroquine, retinoids, or belimumab. Cytotoxic medications such as methotrexate, azathioprine, or mycophenolate mofetil may also be effective. In therapy-resistant lupus dermatitis there are reports of success with topical tacrolimus (caution must be exerted because of the possible increased risk for malignancies) or with systemic dapsone or thalidomide (the extreme danger of fetal deformities from thalidomide requires permission from and supervision by the supplier; peripheral neuropathy is also common).

PREVENTIVE THERAPIES

Prevention of complications of SLE and its therapy include providing appropriate vaccinations (the administration of influenza and pneumococcal vaccines has been studied in patients with SLE; flare rates are similar to those receiving placebo) and suppressing recurrent urinary tract infections. In patients receiving glucocorticoids, the higher the daily dose the lower the immune response to vaccination; however, the great majority of patients achieve protective levels. Vaccination with attenuated live viruses is generally discouraged in patients who are immunosuppressed; however, a recent study of vaccination of a small number of SLE patient with zostavax showed safety and efficacy. Strategies to prevent osteoporosis should be initiated in most patients likely to require long-term glucocorticoid therapy and/or with other predisposing factors. Postmenopausal women can be partially protected from steroid-induced osteoporosis with calcium supplementation, Vitamin D, and either bisphosphonates or denosumab. Safety of long-term use of these strategies in premenopausal women is not well established. Control of hypertension and appropriate prevention strategies for atherosclerosis, including monitoring and treatment of dyslipidemias, management of hyperglycemia, and management of obesity, are recommended. There is increasing evidence that statin therapies can reduce deaths from cardiac events in SLE patients. Finally, the physician must keep in mind that some cancers are increased in SLE patients including non-Hodgkin lymphomas and cancers of thyroid, lung, liver, and vulvar/vaginal tissues.

EXPERIMENTAL THERAPIES

Studies of highly targeted experimental therapies for SLE are in progress. They include (1) inhibition of IFN-α, which was promising in phase II clinical trials, (2) inhibition of IL12 and IL23 signaling; (3) inhibition of IL17; (4) inhibition of IL-6; (5) elimination of plasma cells; (6) inhibition of B/T cell second signal coactivation with CTLA-Ig or anti-CD40L; (7) inhibition of innate immune activation via TLR7 or TLR7 and 9; (8) induction of regulatory T cells with peptides from immunoglobulins or autoantigens or with low doses if IL2; and (9) inhibition of lymphocyte activation by blockade of Jak/Stat. A few studies have used vigorous untargeted immunosuppression with high-dose cyclophosphamide plus anti-T cell strategies, with rescue by transplantation of autologous hematopoietic stem cells for the treatment of severe and refractory SLE. One U.S. report showed an estimated mortality rate over 5 years of 15% and sustained remission in 50%. It is hoped that in the next edition of this text, we will be able to recommend more effective and less toxic approaches to treatment of SLE based on some of these strategies.

PATIENT OUTCOMES, PROGNOSIS, AND SURVIVAL



Survival in patients with SLE in the United States, Canada, Europe, and China is ~95% at 5 years, 90% at 10 years, and 78% at 20 years. In the United States, African Americans and Hispanic Americans with a mestizo heritage have a worse prognosis than whites, whereas Africans in Africa and Hispanic Americans with a Puerto Rican origin do not. The relative importance of gene mixtures and environmental differences accounting for ethnic differences is not known. Poor prognosis (~50% mortality in 10 years) in most series is associated with (at the time of diagnosis) high serum creatinine levels (>124 μmol/L [>1.4 mg/dL]), hypertension, nephrotic syndrome (24-h urine protein excretion >2.6 g), anemia (hemoglobin <124 g/L [<12.4 g/dL]), hypoalbuminemia, hypocomplementemia, antiphospholipid antibodies, male sex, ethnicity (African American, Hispanic with mestizo heritage), and low socioeconomic status. Data regarding outcomes in SLE patients with renal transplants show mixed results: some series show a twofold increase in graft rejection compared to patients with other causes of ESRD, whereas others show no differences. Overall patient survival is comparable (85% at 2 years). Lupus nephritis occurs in ~5% of transplanted kidneys. Disability in patients with SLE is common due primarily to chronic fatigue, arthritis, and pain, as well as renal disease. As many as 30–50% of patients may achieve low disease activity (defined as mild activity on hydroxychloroquine with or without low dose glucocorticoids); fewer than 10% experience remissions (defined as no disease activity on no medications). Both of these conditions may persist for a few years, but are usually not permanent, as flares of SLE occur. The leading causes of death in the first decade of disease are systemic disease activity, renal failure, and infections; subsequently, thromboembolic events become increasingly frequent causes of mortality.

DRUG-INDUCED LUPUS

This is a syndrome of positive ANA associated with symptoms such as fever, malaise, arthritis or intense arthralgias/myalgias, serositis, and/or rash. The syndrome appears during therapy with certain medications and biologic agents, is predominant in whites, has less female predilection than SLE, rarely involves kidneys or brain, is rarely associated with anti-dsDNA, is commonly associated with antibodies to histones, and usually resolves over several weeks after discontinuation of the offending medication. The list of substances that can induce lupus-like disease is long. Among the most frequent are the antiarrhythmics procainamide, disopyramide, and propafenone; the anti-hypertensive hydralazine; several angiotensin-converting enzyme inhibitors and beta blockers; the antithyroid propylthiouracil; the antipsychotics chlorpromazine and lithium; the anticonvulsants carbamazepine and phenytoin; the antibiotics isoniazid, minocycline, and nitrofurantoin (Macrodantin); the antirheumatic sulfasalazine; the diuretic hydrochlorothiazide; the antihyperlipidemics lovastatin and simvastatin. Biologics that can cause drug-induced lupus (DIL) include inhibitors

of IFNs and TNF. In DIL, ANA usually appears before symptoms; however, many of the medications mentioned above induce ANA in patients who never develop symptoms of drug-induced lupus. It is appropriate to test for ANA at the first hint of relevant symptoms and to use test results to help decide whether to withdraw the suspect agent.

FURTHER READING

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350 Antiphospholipid Syndrome

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DEFINITIONS

Antiphospholipid syndrome (APS) is an autoantibody-mediated acquired thrombophilia characterized by recurrent arterial or venous thrombosis and/or pregnancy morbidity. It affects primarily females. APS may occur alone (primary) or in association with other autoimmune diseases, mainly systemic lupus erythematosus (SLE) (secondary). Catastrophic APS (CAPS) is a life-threatening rapidly progressive thromboembolic disease involving simultaneously three or more organs.

The major autoantibodies detected in patients' sera are directed against phospholipids and/or phospholipid (PL)-binding plasma proteins such as prothrombin and β2 glycoprotein I (β2GPI). PLs are components of the cytoplasmic membrane of all living cells. The antibodies are directed against negatively charged PLs including among others cardiolipin, phosphocholine, and phosphatidylserine. The plasma protein β2GPI is a 43-kDa plasma apolipoprotein, which consists of 326 amino acids arranged in five domains (I through V). Domain V forms a positively charged patch, suitable to interact with negatively charged PLs. In plasma, β2GPI has a circular conformation with domain V binding to and concealing the B cell epitopes lying on domain I. Another group of antibodies termed *lupus anticoagulant* (LA) prolongs clotting times in vitro, which are not corrected by adding normal plasma **(Table 350-1)**. Patients with APS often possess antibodies recognizing *Treponema pallidum* PL/cholesterol complexes, detected by Venereal Disease Research Laboratory (VDRL) tests and characterized as biologic false-positive serologic tests for syphilis (BFP-STS).

TABLE 350-1 Classification and Nomenclature of Antiphospholipid Antibodies

NAME	ASSAY FOR THEIR DETECTION	COMMENTS
Antibodies against cardiolipin (aCL)	Enzyme-linked immunosorbent assay (ELISA) using as antigen cardiolipin (CL), a negatively charged phospholipid	aCL from patients with APS recognize β2GPI existing in the human serum as well as in bovine serum, which is used to block the nonspecific binding sites on the ELISA plate. CL simply stabilizes β2GPI at high concentration on the polystyrene surface.
Antibodies against β2GPI (anti-β2GPI)	ELISA using as antigen affinity purified or recombinant β2GPI in the absence of PL	Antibodies recognize β2GPI bound in the absence of CL to an oxidized polystyrene surface, where oxygen atoms in the moieties C–O or C=O were introduced by γ-irradiation.
Lupus anticoagulant (LA)	Activated partial thromboplastin time (aPTT) Kaolin clotting time (KCT) Dilute Russel viper venom test (DRVVT)	Antibodies recognize β2GPI or prothrombin (PT) and elongate aPTT, implying that they interfere with the generation of thrombin by prothrombin. Prolongation of the clotting times is an in vitro phenomenon, and LA induces thromboses in vivo.

Abbreviations: APL, antiphospholipid syndrome; β2GPI, β2 glycoprotein I; PL, phospholipid.

EPIDEMIOLOGY

The incidence of APS is estimated to be around 5 cases per 100,000 persons per year. Anti-PL antibodies occur in 1–5% of the general population. Their prevalence increases with age; however, it is questionable whether they are able to induce thrombotic events in elderly individuals. Moreover, one-third of patients with SLE and other autoimmune diseases **(Chap. 349)** possess these antibodies, with only 5–10% of them developing APS.

PATHOGENESIS

The initiating event for the induction of antibodies to PL-binding proteins seems to be infections, oxidative stress, and major physical stresses such as surgery or trauma. All these factors appear to induce increased apoptosis of the vessel endothelial cells and subsequent exposure of PLs. The latter, bound with serum proteins such as β2GPI or prothrombin, lead to neoantigen formation, which in turn triggers the induction of anti-PLs. The binding of anti-PLs to the disrupted endothelial cells leads to initiation of intravascular coagulation and thrombus formation. Complement activation has been also proposed as a mechanism of APS-related fetal injury.

CLINICAL MANIFESTATIONS AND LABORATORY FINDINGS

Clinical manifestations represent the consequences of venous or arterial thrombosis and/or pregnancy morbidity **(Table 350-2)**. Venous thrombosis, superficial or deep, occurs primarily in the lower extremities often leading to pulmonary emboli. Thrombosis of the pulmonary arteries leads to pulmonary hypertension and thrombosis of the inferior vena cava to Budd-Chiari syndrome. Cerebral venous thrombosis, presents with signs and symptoms of intracranial hypertension and retinal vein thrombosis. Arterial thrombosis affects more commonly the arteries of the brain and is manifested as migraines, cognitive dysfunction, transient ischemic attacks, stroke, and retinal artery occlusion. Arterial thrombosis of the extremities presents with ischemic leg ulcers, digital gangrene, avascular bone necrosis, while thrombosis of other arteries leads to myocardial infarction, renal artery stenosis, glomerular lesions, and infarcts of spleen, pancreas, and adrenals.

Livedo reticularis consists of a mottled reticular vascular pattern that appears as a lace-like, purplish discoloration of the skin. It is probably caused by swelling of the venules due to obstruction of capillaries by thrombi. This clinical manifestation usually occurs together with vascular lesions in the central nervous system and with aseptic bone necrosis. Libman-Sacks endocarditis consists of very small vegetations, histologically characterized by organized platelet-fibrin microthrombi surrounded by growing fibroblasts and macrophages. Glomerular involvement is manifested with hypertension, mildly elevated serum creatinine levels, as well as mild proteinuria/hematuria. Histologically, in an acute phase, thrombotic microangiopathy is present in the glomerular capillaries. In a chronic phase, fibrous intima hyperplasia, fibrous and/or fibrocellular arteriolar occlusions, and focal cortical atrophy are present (Table 350-2). Pregnancy morbidity manifests with increased risk of recurrent miscarriages, intrauterine growth